



A hands-on tutorial



Playwright-style mobile test automation on Appium 3

Deck · 12 labs · demo app included

Test the **Playwright way** on top of Appium 3 — walking the **entire taqwright docs**, top to bottom, with a runnable lab after each topic.



What's in this repo



The repo at a glance

 <code>slides.md</code>	this deck — the docs, top to bottom (<code>slides.html</code> / <code>slides.pdf</code> built alongside)
 12 lab folders	one runnable project per topic, each with its own <code>README.md</code>
 <code>app/</code>	the demo-app binaries under test — committed , nothing to download
 <code>images/</code>	diagrams used by the deck

Clone it and go. Every lab is a self-contained TS project: `tests/*.spec.ts` + `taqwright.config.ts` .



The app under test

- 📦 `taqelah/demo-app v1.0.0` — APK + `.app`, committed under `app/`
- 🔑 Credentials: `emma@demoapp.com` / `10203040`
- 🤖 Appium 3.x · UiAutomator2 (Android) · XCUITest (iOS, macOS only)

Every lab's `buildPath` already points at `app/` — no downloads, no per-lab setup. Install by hand if you like: `adb install -r app/DemoApp-v1.0.0.apk`



Why taqwright?

Raw Appium/WDIO is powerful but **boilerplate-heavy**: explicit waits, session setup, brittle selectors.

- 🥰 Playwright's test runner + a flat `mobile` locator API
- ⌚ Auto-waiting locators · 🔁 auto-retrying `expect()` — far less flake
- 📱 iOS + Android from the same test
- 📺 Record a flow · 🐛 debug mode

If you know Playwright for web, you already know 80% of this.



What it is

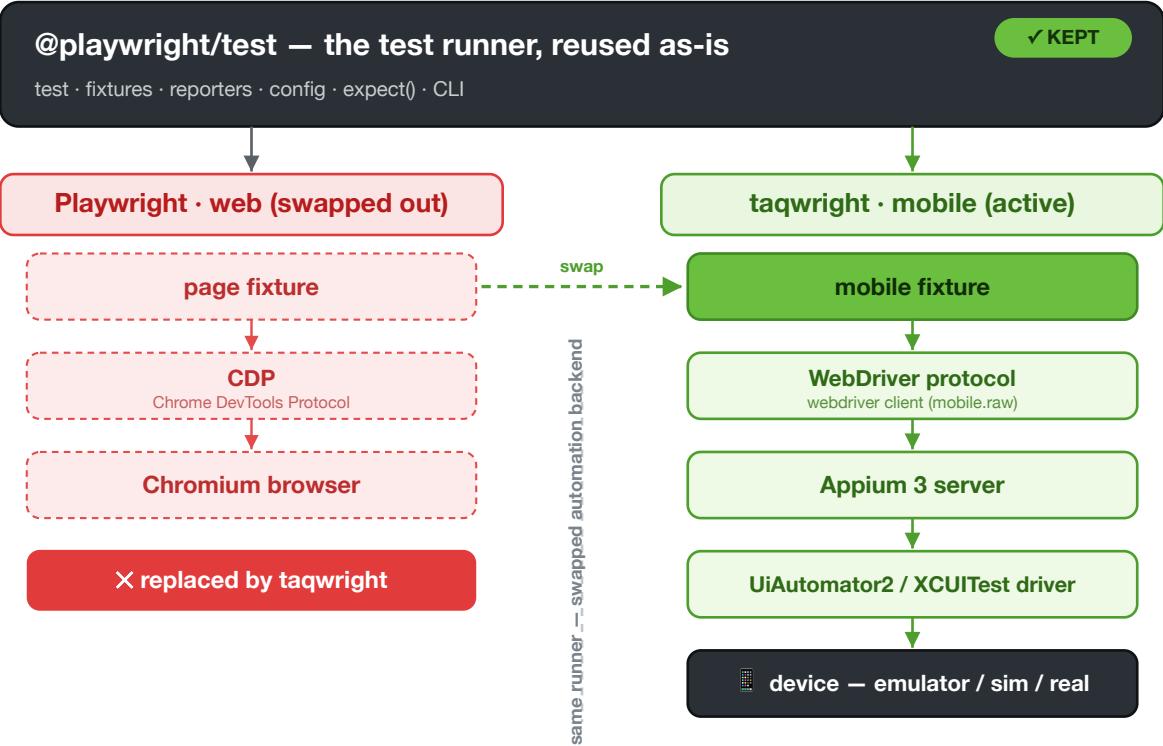
taqwright = the Playwright test runner with a flat **mobile** locator API on top of Appium 3.

- `import { test, expect } from 'taqwright'`
- The **mobile** fixture: `getById`, `getByLabel`, `click`, `fill`, `expect(...).toBeVisible()`
- **TypeScript only** — `tests/*.spec.ts` + `taqwright.config.ts`

⚠ No `node/` + `python/` dual-stack. One TS project per lab — that's the whole point.



Architecture — it overrides Playwright with Appium



Built on `@playwright/test ^1.60 + webdriver ^9.27` · not a fork — same runner, Appium backend.
Escape hatch: `mobile.raw` → the underlying WebDriver client.

Keep the Playwright runner · **swap** the backend: `mobile` → WebDriver → Appium 3.



Under the hood — reused vs swapped

Layer	Playwright (web)	taqwright (mobile)
Runner · fixtures · reporters · expect	@playwright/test	same — reused as-is
Automation fixture	page	mobile
Protocol	CDP (Chrome DevTools)	WebDriver (webdriver pkg)
Backend	Chromium	Appium 3 → UiAutomator2 / XCUITest
Escape hatch	page internals	mobile.raw (raw WebDriver)

Not a fork. taqwright keeps the entire Playwright test experience and only swaps the bottom of the stack — so if you know Playwright, you already know taqwright. Built on `@playwright/test ^1.60` + `webdriver ^9.27`.



How this deck maps to the docs

The deck walks the **taqwright docs sidebar, top to bottom** — two groups:

Part 1 · Getting started

1. Installation
2. Codegen
3. Writing tests
4. Running & debugging

Part 2 · Taqwright Test

Actions · Assertions · Auto-waiting ·
Configuration · Annotations · Global setup &
teardown · Parallelism · Parameterize · Projects ·
Reporters · Retries · Sharding · Timeouts · Trace
viewer



The labs — hands-on after each topic (1 / 2)

Lab	Exercises	After	Status
 tutorial_01_setup_lab/	install, provision & verify the toolchain	Installation	
 tutorial_02_codegen_lab/	record a flow into a spec	Codegen	 scaffold
 tutorial_03_config_lab/	6 mini-projects — one per config concept	Configuration	
 tutorial_04_annotations_lab/	tags · metadata · skip/fixme/fail/slow	Annotations	
 tutorial_05_global_setup_lab/	globalSetup/teardown module + setup project	Global setup	
 tutorial_06_parallel_lab/	workers · device pool · autoDiscover	Parallelism	

Each lab appears right after the topic it exercises — not bunched at the end. **Start with `tutorial_01_setup_lab`** . Every folder has its own `README.md` with the full walkthrough.



The labs (2 / 2)

Lab	Exercises	After	Status
 tutorial_07_parameterize_lab/	data-array loop · external JSON	Parameterize	
 tutorial_08_projects_lab/	android + iOS targets via <code>projects[]</code>	Projects	 android
 tutorial_09_merge_report_lab/	blob reporter → <code>merge-reports</code>	Reporters	
 tutorial_10_retry_timeout_trace_lab/	flaky+retry · timeouts · trace	§17–19	
 tutorial_11_page_object_lab/	one spec, both platforms (base + subclasses)	Page Objects	
 tutorial_12_cloud_debug_lab/	page-object spec on BrowserStack	last topic	 android

 = runnable on a local emulator/sim; cloud + iOS need creds / macOS.



Part 1 · Getting started 🚀



1 • Installation

```
node -v                                # ⚠ must be v24+ – the ONE hard requirement
nvm install 24 && nvm use 24 && nvm alias default 24

npm init taqwright                     # scaffold a new project (config, tsconfig, example spec)
# ...or add taqwright to an existing project:
npm i -D @taqwright/taqwright
```

`taqwright doctor` **errors** below Node 24; every other check is a soft warn. 📦 Package is `@taqwright/taqwright` — our labs alias it to `taqwright` so `import ... from 'taqwright'` + `npx taqwright` both work.



1 · Installation — watch (3 min)



► Taqwright in 3 minutes

install, scaffold, and run your first Android test

TAQELAH Singapore

Tap to watch the whole getting-started loop end to end — then we'll break down exactly what `init` asks and does.



1. **taqwright init** — the interactive flow

`npm init taqwright` asks a short series of questions (Enter accepts the (default)):

1. Project location (`./taqwright-tests`)
2. *(if not empty)* continue and write into it?
3. Test folder name (`tests`)
4. Platform [`android/ios/both`] (*ios/both* = *macOS only*)
5. **Android** → **probes your machine** → Auto-install the Android toolchain? (~700 MB)
6. Also create an emulator? (~1 GB) (*or pick an existing AVD*)
7. Download the demo app? · Run `npm install`?

How you answer: text = type or Enter · choice = type `android / ios / both` · confirm = `y / n` (capital = default) · AVD = a number. Bad input just re-asks.



1. **taqwright init** — a sample run

`taqwright init` — scaffold a new project

Android toolchain:

- ✓ JDK (java v25.0.2)
- ✓ Android SDK (adb)
- ✓ Appium 3.x (v3.3.1)
- ✓ uiautomator2 driver
- ✓ Android emulator (AVD): Pixel_7_API_34
- detected a working Android toolchain — skipping install.

Created:

<code>my-app/package.json</code>	<code>my-app/taqwright.config.ts</code>
<code>my-app/tsconfig.json</code>	<code>my-app/tests/example.spec.ts</code>
<code>my-app/.npmrc</code>	<code>my-app/.gitignore</code>

Next steps:

```
cd my-app
npx taqwright test
```

Commands:

```
npx taqwright doctor
npx taqwright codegen · test · show-report
```



1. **taqwright init** — outcomes, flags & CI

Branches & outcomes

- toolchain **detected** → **skips** install
- **fresh** → installs (~700 MB) ± **AVD** (~1 GB)
- demo app added **vs** no-op stub
- **!** aborts: Node < 24 · **iOS on non-mac** · non-empty dir (no **--yes**)
- npm / toolchain fail → *scaffold still succeeds*, retry later

Flags / non-interactive

```
npx taqwright init my-app \  
  --platform android \  
  --test-dir tests \  
  --no-demo-app --yes
```

All-flags or no-TTY → runs **scripted** with defaults (great for CI). Toggle prompts with **--[no-]install** · **--[no-]install-toolchain** · **--[no-]with-avd** · **--[no-]demo-app**.

Then → **doctor** (ready?) · **devices** (targets?) · **install** (provision).



1. **taqwright doctor** — is my machine ready?

A **pre-flight check** of the whole mobile toolchain before you run a test.

```
taqwright doctor (v1.0.0)
[ok] Node.js 24+                - v24.15.0
[ok] adb (Android SDK)          - on PATH
[ok] ANDROID_HOME                - ../Library/Android/sdk
[ok] java / JAVA_HOME (UiA2)    - v25.0.2
[ok] Appium (test server)       - v3.3.1
[ok] Appium drivers              - uiautomator2, xcuitest
[ok] xcrun / Xcode (XCUITest)   - Xcode 26.5
```

13 checks across Android + iOS. **Node 24 is the only hard fail** — everything else is a soft **warn**, and you only need the rows for *your* platform. `--json` for CI / agents. Run it first whenever something's off.



1. taqwright devices — what can I target?

Lists every target taqwright can attach to — Android AVDs, iOS simulators, connected devices.

```
Android (adb + emulator):          ← example output; YOUR AVD names differ
  Pixel 7 API 34                    avd:Pixel_7_API_34    (shutdown)
  taqwright api34                   avd:taqwright_api34   (shutdown)

iOS Simulators (xcrun simctl):
  iPhone 17 Pro                     5BE0...1F9EA698    (shutdown, iOS 26.5)
  iPhone Air                        FF95...A16985F      (booted,    iOS 26.5)
```

📋 Copy the `avd:<id>` into `device.name` (Android) or the `<UDID>` into `device.udid` (iOS) in `taqwright.config.ts`.

Never copy an AVD name off a slide or someone else's config — always take it from *your* devices output. State = shutdown / booted — taqwright can cold-boot for you via `appium.autoStartDevice` or `device.autoDiscover`.



1. `taqwright install` — provision the toolchain

Zero-touch setup on a fresh machine — installs **JDK + Android SDK + Appium** (and the drivers).

```
npx taqwright install          # JDK + Android SDK + Appium — zero-touch
npx taqwright install --with-avd # ...also build a system image + emulator (~1 GB)
npx taqwright install --force   # reinstall even if already provisioned
npx taqwright install --print-env # print export lines to use the toolchain from your shell
```

⚠ **Android only** — iOS needs **Xcode** (manual). Sizes: ~700 MB toolchain, +~1 GB with the AVD.

💡 **Prefer your own SDK.** `install` sets up a **separate, taqwright-managed** toolchain — handy for a *quick test* or a brand-new laptop, but it **won't see or drive an SDK / emulator you already have**. **Already set up? Skip `install`** and just point `device.name` at your existing AVD.



Lab · `tutorial_01_setup_lab` get your machine ready

One-time **environment setup** before any taqwright test — end on a green smoke run.

```
cd tutorial_01_setup_lab      # the install & setup lab
npm use 24                   # taqwright needs Node 24+ (the one hard requirement)
npm install                  # installs taqwright

npx taqwright doctor          # verify the toolchain – 13 checks, all green
npx taqwright test tests/smoke.spec.ts # run the smoke spec – boots a device, reinstalls the app, asserts login
npx taqwright show-report     # open the HTML report
```

No setup beyond this: the config `autoDiscover` s + boots a device, auto-starts Appium, and reinstalls the shared `app/` APK each test. Green here = your machine's ready for every lab in this repo. → [tutorial_01_setup_lab/](#)



2 · Codegen

Playwright-style codegen for mobile

Record a flow on a live device → a clean taqwright spec. Deep dive into the inspector.



See it in action



► Taqwright Codegen

Playwright-style codegen for mobile

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Tap to watch: connect a device, drive the app, and watch a runnable spec build itself — locators ranked live.

[🔗 youtube.com/watch?v=FBxFg1XbuKE](https://youtube.com/watch?v=FBxFg1XbuKE)



2 · Codegen — launch & connect

```
npx taqwright codegen
```

- Opens a **web inspector** at `localhost:4280` (auto-picks a free port) in your browser
- Reads `taqwright.config.ts` — project, caps (UiAutomator2, `noReset`), Appium host/port
- Connect to a **local** emulator/device (*Appium auto-starts*) or a **cloud** device (BrowserStack / LambdaTest)
- **Boot / stop** emulators & simulators right from the UI · Ctrl+C cleans the session up



2 · Codegen — on a cloud device ☁️

No local emulator? Point the inspector at a **real device on BrowserStack** (or LambdaTest) and still get the **live screen + inspect + record**.

```
export BROWSERSTACK_USERNAME=...      # inspector pre-fills these
export BROWSERSTACK_ACCESS_KEY=...
npx taqwright codegen
```

- ① creds from env (or type them) → ② pick a **device + OS** from the listed cloud devices → ③ point at your uploaded app (`bs://...`)
- Same **live screen mirror**, same **ranked + uniqueness-checked** locators, same **recording** → **spec**
- Real **permission dialogs** show up (auto-grant is off during codegen) — so you can record them

Generate on a cloud device, then run the spec on the same grid → [tutorial_12_cloud_debug_lab/](#).



2 · Codegen — the inspector UI

Four things on one screen:

-  **Live screen mirror** — tap / type / swipe the real device in your browser
-  **View tree** — the live accessibility tree (never guess from a screenshot)
-  **Locators tab** — every viable locator, **ranked + uniqueness-checked** against the screen
-  **Recorded script tab** — ready-to-paste taqwright code, filling in as you act

Native ↔ webview? The inspector lists contexts and **switches** between them.



2 · Codegen — the locator suggester

For the selected element it generates candidates **per category**, **verifies each against the live screen** (how many match · is it unique), then **recommends** the best.

Priority order (*from source*)

Android: `getById` → `getByUiSelector` → `getByXpath`

iOS: `getById` → `getByPredicate` → `getClassChain` → `getByXpath`

Web view: `getByCss` → `getByXpath`

Same ranking as `taqwright inspect` — codegen just also records what you do with it.

Smarts

-  prefers **unique** + stable (id) over positional
-  not unique? falls back to `.nth(i)`
-  flags **positional** locators as fragile



2 · Codegen — what it records · actions

Every gesture and action becomes a typed step (then `toSpec()` → code):

Gesture / screen

`tap` · `swipe` · `screenScroll`

Touch

`click` · `doubleTap` · `longPress` · `press` ·

`focus` · `blur`

Input & controls

`fill` · `clear` · `pressSequentially` · `check`
/ `unchecked` · `selectOption` (label / index / date
/ time)

Movement

`swipe` · `scrollIntoView` · `pinch` (in/out) ·
`dragTo` (target)

Plus `sendKeys` (device keyboard), `switchContext` (native↔webview), and freeform `comment` lines.



2 · Codegen — what it records · assertions

Click "assert" on an element → an **auto-retrying** `expect(...)` is recorded:

State

`toBeVisible` / `toBeHidden`

`toBeEnabled` / `toBeDisabled`

`toBeChecked` / `unchecked`

`toBeEditable` / `readonly`

`toBeFocused` · `toBeAttached`

`toBeEmpty` · `toBeInViewport`

Value

`toHaveText` (exact | contains)

`toHaveValue`

`toHaveCount`

`toHaveAttribute`

Recorded assertions are real auto-retrying matchers — not one-shot `isVisible()` reads.



2 · Codegen — output & export

`Recorder.toSpec()` writes a runnable **taqwright TypeScript** spec — but it's not the only target:

```
// recorded.spec.ts (taqwright)
test('flow', async ({ mobile }) => {
  await mobile.getById('Login').click();
  await expect(mobile.getByText('Welcome'))
    .toBeVisible();
});
```

Also exports the same recording as:

-  **Python** Appium steps
(`toStepsPython`)
-  **Java** Appium steps (`toStepsJava`)

Then **clean it up**: swap any positional/xpath locators for the recommended stable one, keep the assertions. Codegen is a *starting point*, not the finished test.



Lab · `tutorial_02_codegen_lab` record a flow

Stop hand-writing locators — **record** them: the inspector ranks candidates by **stability + uniqueness** and emits paste-ready taqwright code.

This lab ships with no spec — `tests/` is empty. Record the *add to cart* flow on the demo app, save it into `tests/`, then run it.

```
cd tutorial_02_codegen_lab    # the codegen / record lab — no spec ships with it
npm install                  # installs taqwright

npx taqwright codegen        # record on the demo app: log in → Shop All → search "black dress" → add to cart
mkdir -p tests               # paste the Recorded script into tests/search-add-to-cart.spec.ts
npx taqwright test tests/search-add-to-cart.spec.ts # run the spec you just recorded
npx taqwright show-report    # open the HTML report
```

Clean it up: swap positional/xpath for the recommended stable locator, keep the assertions. Captures actions · gestures · assertions (→ retrying `expect()`); can export **Python / Java** too.  Watch the codegen video first.



3 · Writing tests

```
import { test, expect } from 'taqwright'; // docs: '@taqwright/taqwright'

test('a valid user can log in', async ({ mobile }) => {
  // login fields are Flutter EditTexts with no label → match the hint
  await mobile.getByXpath("//*[@hint='Username']").fill('emma@demoapp.com');
  await mobile.getByXpath("//*[@hint='Password']").fill('10203040');
  await mobile.getByLabel('Login').click(); // button has an accessibility id
  await expect(mobile.getByLabel('View All')).toBeVisible();
});
```

- Each test gets a **fresh Appium session**, torn down automatically
- Locators are **lazy** — nothing runs until an action or assertion
- Actions **auto-wait** for visible + enabled — **no sleeps**



3 · Writing tests — locator getters

Getter	Resolves to	Stability
<code>getById</code> / <code>getById</code>	resource-id / accessibility id	★ ★ ★ best
<code>getByLabel</code>	accessibility label / content-desc	★ ★ ★
<code>getByRole('button', { name })</code>	widget-type mapping	★ ★
<code>getByText(/.../i)</code>	visible text	★ fragile (i18n)
<code>getByType('android.widget....')</code>	class / XCUI type	last resort
<code>getByXPath</code>	raw XPath	🐢 slowest

Demo-app login fields are Flutter `EditText` s with no id → `getByType('android.widget.EditText').nth(0|1)` . Button & home marker have ids → `getByLabel` .



3 · Writing tests — fresh state

```
// Option A — relaunch in a hook (no buildPath needed)
test.beforeEach(async ({ mobile }) => {
  await mobile.terminateApp().catch(() => {});
  await mobile.launchApp();
});
```

```
// Option B — reinstall before every test (config)
resetBetweenTests: true,
buildPath: './DemoApp-v1.0.0.apk',
appBundleId: 'com.taqelah.demo_app',
```

Default: sessions persist across tests in a worker. Keep `beforeEach` **light** — it runs every test.



4 · Running & debugging

```
npx taqwright test                # all projects (Appium auto-starts)
npx taqwright test login.spec.ts  # one file
npx taqwright test --grep "log in" # by title / tag
npx taqwright test --project android # one project
npx taqwright show-report         # open the HTML report
```

- 📸 **trace** · 🎥 **video** · 🌐 **network** — each `off` / `on` / `on-failure`
- Artifacts land in `test-results/` (default `outputDir`); taqwright prints the exact `show-report` command at the end
- 🐛 `await mobile.pause()` hands the running session to the inspector



Part 2 · Taqwright Test ⚙️



6 · Actions — on a locator

```
await mobile.getByXPath("//*[@hint='Username']").fill('emma@demoapp.com');  
await mobile.getByLabel('darkMode').check();  
await mobile.getByType('android.widget.Spinner').selectOption('Large');
```

- **Text:** fill · clear · pressSequentially
- **Controls:** check · uncheck · selectOption (picker / spinner / date-time)
- **Touch:** click / tap · doubleClick · longPress · press · focus · blur
- **Gestures:** swipeLeft/Right/Up/Down · dragTo · pinchIn / pinchOut · scrollIntoView

Every action **auto-waits** for visible + enabled first. No web-only file-upload / hover on mobile.



6 · Actions — screen & lifecycle

```
await mobile.swipe('up');           // whole-screen
await mobile.scrollToView(locator);
await mobile.pressButton('BACK');   // BACK · HOME · POWER · VOLUME_*
await mobile.launchApp();           // app lifecycle
await mobile.terminateApp();
```

- **Screen:** swipe · scroll · scrollToView · pressButton · goBack · screenshot
- **Lifecycle:** launchApp · terminateApp · activateApp · installApp · openDeepLink
- **OS:** setOrientation · hideKeyboard · acceptAlert / dismissAlert · setLocation · clipboard · permissions

Escape hatch: `mobile.get raw` → the underlying WebDriver.



7 · Assertions — auto-retrying `expect()`

```
await expect(mobile.getByLabel('Submit')).toBeEnabled();
await expect(mobile.getByText(/welcome/i)).toBeVisible({ timeout: 10_000 });
await expect(mobile.getByText('Error')).not.toBeVisible();
```

State: `toBeVisible` · `toBeHidden` · `toBeEnabled` · `toBeDisabled` · `toBeChecked` ·

`toBeEditable` · `toBeFocused` · `toBeAttached` · `toBeInViewport` · `toBeEmpty`

Value: `toHaveText` · `toContainText` · `toHaveValue` · `toHaveCount` · `toHaveAttribute`

Auto-retries up to 30 s. Two styles: `expect(locator).toBe...` or `locator.assert...`.



7 · Assertions — beyond the basics

```
expect.soft(mobile.getByText('A')).toBeVisible(); // record, don't stop
await expect(mobile.getByText('B')).toBeVisible(); // run keeps going

await expect.poll(() => api.status()).toBe('ready'); // poll a value
await expect(async () => { /* ... */ }).toPass();      // retry a block
```

- `.not` works on **state** matchers (value matchers reject it — clear error)
- Plain values (`expect(2+2).toBe(4)`) fall through to **non-retrying** native assertions
- Second arg = a **custom message** shown in the report

⚠ Web-only Playwright matchers (`toHaveCSS` , `toHaveClass` ...) **throw**.



Appium-by-hand vs taqwright

Raw Appium — WebdriverIO

```
const u = await $('android=new ' +
  'UiSelector().className(' +
  '"android.widget.EditText"' +
  '.instance(0)')
await u.waitForDisplayed({ timeout: 20000 })
await u.setValue('emma@demoapp.com')
// ...password, click...
expect(await $('~View All')
  .isDisplayed()).toBe(true)
```

taqwright

```
await mobile.getByType(
  'android.widget.EditText')
  .nth(0).fill('emma@demoapp.com')
// ...password, click...
await expect(
  mobile.getLabel('View All'))
  .toBeVisible()
```

Same flow, same driver underneath — **fewer lines, auto-waits built in, one language.**



8 · Auto-waiting

Before driving an element, taqwright runs **actionability checks** and retries them, polling every **200 ms** up to the timeout (30 s).

Actions wait for

✓ visible + enabled

click · fill · check · swipe · drag

Queries wait for

✓ visible only

getText · getValue · boundingBox ·
screenshot

So tapping a button still animating in, or filling a field not yet rendered, **just works** — no sleeps. Need a custom wait?

```
locator.waitFor({ state: 'hidden' | 'attached' | 'disabled' }) .
```



8 · Auto-waiting — the trap to avoid

✗ One-shot, racy

```
await mobile.waitForTimeout(2000);  
if (await loc.isVisible()) {  
  // checks ONCE, right now  
}
```

✓ Auto-retrying

```
await expect(loc).toBeVisible();  
// polls until visible  
// or times out
```

`isVisible()` is a boolean at **one instant**; `expect(...).toBeVisible()` **waits**. Hardcoded sleeps are the #1 cause of flake — and unnecessary here.



9 · Configuration — what & where

```
import { defineConfig, Platform } from 'taqwright';

export default defineConfig({
  testDir: './tests',
  timeout: 60_000,
  expectTimeout: 30_000,
  reporter: [['list'], ['html', { open: 'never' }]],
  projects: [ /* one per device target – each has its own `use` */ ],
});
```

One file (`taqwright.config.ts`), generated by `init`. **Top-level keys = the whole run**; `projects[]` = where/how each **device target runs** — and a project key **overrides** the top-level one.

🔧 Each concept below has a hands-on **mini-project** in [tutorial_03_config_lab/](#) — try it right after the slide.



9 · Configuration — top-level · tests & timing

Key	Type	Default	What it does
testDir	string	.	Root folder taqwright scans for specs
testMatch	string RegExp []	**/*.spec.ts	Which files are tests
testIgnore	string RegExp []	—	Files to skip
timeout	number (ms)	30_000	Per-test budget (hooks included)
expectTimeout	number (ms)	5_000	Bounds assertions + actions ; polls every 200 ms
retries	number	0	Re-run a failed test from scratch (<i>deep dive: §17</i>)

All of these are **overridable per project** and via CLI flags (`--timeout` , `--retries` , ...).



Lab · `tutorial_03_config_lab/1_tests_and_timing/`

```
cd tutorial_03_config_lab/1_tests_and_timing  
npm install && npm test
```

Spotlights `testDir` · `timeout` · `expectTimeout` · `retries`. Try: drop `timeout` to `5_000`; break a locator with `retries: 1` → watch the rerun + **flaky** flag. → [1_tests_and_timing/](#)



9 · Configuration — top-level · execution & output

Key	Type	Default	What it does
workers	number	1	Parallel device workers (§12)
fullyParallel	boolean	false	Split tests within a file across workers
reporter	name [name, opts] []	'list'	list · line · dot · html · json · junit · blob · github (§15)
outputDir	string	test- results	Where traces / videos / artifacts land
globalSetup	string string[]	—	Module run once before all tests (§11)
globalTeardown	string string[]	—	Module run once after all tests
forbidOnly	boolean	false	Fail the run if a stray test.only is committed (CI)



Lab · **tutorial_03_config_lab/2_execution_and_output/**

```
cd tutorial_03_config_lab/2_execution_and_output  
npm install && npm test
```

Spotlights `workers` · `reporter` · `globalSetup` · `forbidOnly` . Watch the `[global-setup]` log, `report.json` , and `./artifacts` . Try: `workers: 2` ; add `test.only` → `forbidOnly` fails the run. → [2_execution_and_output/](#)



9 · Configuration — the `use` block · app & session

Each project's `use` describes **one device target + how the app is handled**:

Key	Type	Notes
<code>platform</code>	<code>Platform.ANDROID</code> <code>.IOS</code>	required
<code>device</code>	<code>DeviceConfig</code>	required — see <i>device providers</i> →
<code>appBundleId</code>	<code>string</code>	App package id (<code>com.taqelah.demo_app</code>)
<code>buildPath</code>	<code>string</code>	Path to <code>.apk</code> / <code>.app</code> (rel. to config)
<code>resetBetweenTests</code>	<code>boolean</code>	Reinstall + relaunch before every test
<code>capabilities</code>	<code>Record<string, unknown></code>	Extra raw Appium caps, merged in
<code>expectTimeout</code>	<code>number</code> (ms)	Per-project override

⚠ **Type rule:** `resetBetweenTests: true` *requires both* `buildPath` **and** `appBundleId` (TS-enforced union). Omit / `false` → app is left as-is between tests.



Lab · `tutorial_03_config_lab/3_use_app_and_session/`

```
cd tutorial_03_config_lab/3_use_app_and_session  
npm install && npm test
```

Spotlights `platform` · `buildPath` · `resetBetweenTests` · `capabilities` . Try: the **reset trio** — `resetBetweenTests: true` + `buildPath` + `appBundleId` — is type-required *together*; comment out any one and **TypeScript errors** (put it back, error clears). → [3_use_app_and_session/](#)



9 · Configuration — use artifacts · trace · video · network

```
use: { trace: 'on-failure', video: 'on-failure', network: 'off' }
```

Key	Captures	
trace	per-action screenshots + page source timeline	(viewer → §19)
video	screen recording (.mp4)	
network	HTTP traffic (HAR)	⚠️ upcoming / WIP

All share the same modes: 'off' (default) · 'on' · 'on-failure' · 'retain-on-failure' (keep on fail, discard on pass). 'on' is heavy — prefer 'on-failure' in Cl. network is ⚠️ work in progress — trace + video are live today.



Lab · `tutorial_03_config_lab/4_artifacts/`

```
cd tutorial_03_config_lab/4_artifacts  
npm install && npm test && npx taqwright show-report
```

Spotlights `trace` · `video` · `network` (modes `off` · `on` · `on-failure` · `retain-on-failure`). `trace:'on'` here → step through **every** action in the report. → [4_artifacts/](#)



9 · Configuration — **device** providers

Local — **emulator** / **local-device**

```
device: {
  provider: 'emulator',
  name: 'Pixel_7_API_34', // ← EXAMPLE — use yours
  udid: 'emulator-5554', // wins over name
  osVersion: '14',
  orientation: 'portrait', // | 'landscape'
  autoDiscover: true, // default: enumerate+boot
  // pool: [{ udid, name?, osVersion? }] // workers>1
}
```

Cloud — **browserstack** / **lambdatest**

```
device: {
  provider: 'browserstack',
  name: 'Google Pixel 8', // required
  osVersion: '14.0', // required
  orientation: 'portrait',
  enableCameraImageInjection: false,
}
```

Four **provider** s. Local defaults to **autoDiscover: true** (no fixed **name** needed); cloud **requires** **name** + **osVersion** .

Cloud creds come from env (**BROWSERSTACK_USERNAME** / **..._ACCESS_KEY**) → §14.



9 · Configuration — device · how to target

1 · `udid` — pin one exact device

```
device: { provider: 'emulator',
          udid: 'emulator-5554' } // wins over name
```

Copy it from `npx taqwright devices` / `adb devices`.

Use when several devices are attached.

2 · `autoDiscover` — let taqwright pick (default)

```
device: { provider: 'emulator', autoDiscover: true }
```

No name/udid needed — enumerates + boots what's available; finds several when `workers > 1`. What

`tutorial_01_setup_lab` uses.

3 · `pool` — explicit devices for workers

```
workers: 2,
device: { provider: 'emulator', pool: [
  { udid: 'emulator-5554' }, // ← YOUR serials
  { udid: 'emulator-5556' }, //   (npx taqwright devices)
] }
```

Each worker grabs one entry `{ udid, name?, osVersion? }`. Overrides auto-discovery. ⚠ Pool entries are **machine-specific** — AVD ids/serials differ on every laptop; always paste your own.

4 · Cloud — BrowserStack / LambdaTest

```
device: { provider: 'browserstack',
          name: 'Google Pixel 8', osVersion: '14.0' }
```

`name` + `osVersion` required · creds via env · `buildPath` uploaded. Run `--project browserstack`.



Lab · **tutorial_03_config_lab/5_device_providers/**

```
cd tutorial_03_config_lab/5_device_providers  
npm install && npm test
```

Spotlights `emulator` · `udid` · `pool` · `cloud`. Try: swap `autoDiscover` for `udid: 'emulator-5554'`; uncomment the `browserstack` project + creds. Precedence: `udid` > `pool` > `name` > `autoDiscover` . → [5_device_providers/](#)



9 · Configuration — the `appium` server

```
use: { appium: { autoStart: true, host: 'localhost', port: 4723, path: '/' } }
```

Key	Default	What it does
<code>autoStart</code>	<code>true</code>	Spawn Appium on <code>host:port</code> if nothing's listening
<code>autoStartDevice</code>	<code>false</code>	Cold-boot an offline emulator (needs string <code>name</code>)
<code>host</code> · <code>port</code> · <code>path</code>	<code>localhost</code> · <code>4723</code> · <code>/</code>	Where to reach Appium
<code>newCommandTimeout</code>	<code>60</code> (s)	Idle time before Appium kills the session
<code>connectionTimeout</code>	<code>5000</code> (ms)	Wait for the server to respond
<code>logLevel</code>	—	<code>trace</code> · <code>debug</code> · <code>info</code> · <code>warn</code> · <code>error</code> · <code>silent</code>

📁 Live, runnable examples of **all** of the above: [tutorial_01_setup_lab/](#) & the [tutorial_03_config_lab/](#) projects' `taqwright.config.ts` .



Lab · `tutorial_03_config_lab/6_appium_server/`

```
cd tutorial_03_config_lab/6_appium_server  
npm install && npm test
```

Spotlights `autoStart` · `autoStartDevice` · `host` / `port` / `path` · `timeouts` · `logLevel` . Try: `logLevel: 'debug'` to see the Appium handshake; start `appium` yourself → [6_appium_server/](#)



10 · Annotations — control execution

```
test.skip('apple pay', async ({ mobile }) => { /* ... */ }); // never runs
test.skip(isCI, 'flaky on CI'); // conditional
test.fixme('known broken', async ({ mobile }) => { /* ... */ }); // won't run/fail
test.fail('reproduces TAQ-9', async ({ mobile }) => { /* ... */ }); // passes IF it fails
test('huge flow', async ({ mobile }) => { test.slow(); /* 3x timeout */ });
```

All take the conditional (condition, reason) form — platform-specific logic in one test, no duplication. `test.only`

focuses a single test while developing.



10 · Annotations — tags & metadata

```
test('checkout', { tag: '@smoke' }, async ({ mobile }) => { /* ... */ });  
test('payments', { tag: ['@smoke', '@payments'] }, async ({ mobile }) => { /* ... */ });
```

```
npx taqwright test --grep @smoke           # run only smoke  
npx taqwright test --grep-invert @payments # everything except payments
```

```
// freeform metadata → surfaces in the report  
test('issue repro', { annotation: { type: 'issue', description: 'TAQ-482' } }, ...);
```

Runtime: push to `test.info().annotations` when a condition emerges mid-run.



Lab · **tutorial_04_annotations_lab/**

```
cd tutorial_04_annotations_lab
npm install
npx taqwright test --list           # see how each test is reported
npx taqwright test --grep @smoke    # run only @smoke-tagged tests
npm test                           # all: skip/fixme excluded · fail expected · slow tripled
```

Demos `tag` · `annotation` `metadata` · `test.skip` / `fixme` / `fail` / `slow` . Try: `--grep-invert @checkout` ; `CI=1 npx`

`taqwright test` flips the conditional `test.skip(isCI, ...)` . → [tutorial_04_annotations_lab/](#)



11 · Global setup & teardown

Setup project (needs a device)

```
// config
projects: [
  { name: 'setup',
    testMatch: /global\.setup\.ts/ },
  { name: 'android',
    dependencies: ['setup'],
    use: { /* ... */ } },
]
```

```
import { test as setup } from 'taqwright';
setup('sign in once', async ({ mobile }) => { /* ... */ });
```

globalSetup module (no device)

```
// config
globalSetup: './global-setup.ts',
globalTeardown: './global-teardown.ts',
```

```
export default async () => {
  const r = await fetch(SEED_URL);
  process.env.SEED = await r.text();
};
```

API calls / seeding / file work only.

Use **globalSetup** / **globalTeardown** for one-time, **device-free** prep shared by all tests (seed a DB · mint a token · start a mock server), undone once at the end. *Per-test* → **beforeEach** ; *device setup* (sign in) → a **setup project**.



11 · Setup scopes — global setup vs spec hooks

Mechanism	Scope	Device?	Use for
<code>globalSetup</code> / <code>globalTeardown</code>	once per run	✗	seed/reset a DB · mint a token · start a mock server
<code>setup project (dependencies)</code>	once, before dependents	✓	sign in once · grant permissions
<code>beforeAll</code> / <code>afterAll</code>	once per file (or <code>describe</code>)	✓	per-file prep — open a screen once
<code>beforeEach</code> / <code>afterEach</code>	every test	✓	reset state · navigate to a start point

`globalSetup` → `setup project` → | `beforeAll` → (`beforeEach` → TEST → `afterEach`)* → `afterAll` | → `globalTeardown`

`globalSetup` / `globalTeardown` = config-level, **no device**, once for the whole run. `beforeAll` / `beforeEach` = in-worker, **have the mobile device fixture**. (These labs get per-test fresh state from `resetBetweenTests` instead of a `beforeEach` .)



Lab · `tutorial_05_global_setup_lab/`

```
cd tutorial_05_global_setup_lab
npm install
npm test      # watch the order: [globalSetup] → setup project → android tests → [globalTeardown]
```

Demos **every scope**: `globalSetup` / `globalTeardown` module · **setup project** (`dependencies`) · `beforeAll` / `afterAll` · `beforeEach` / `afterEach` (in `hooks.spec.ts`). Run `npm test` and read the order in the console. →

[tutorial_05_global_setup_lab/](#)



12 · Parallelism

taqwright runs **serially by default** (`workers: 1`). Scale out across **devices**:

```
workers: 3,  
use: {  
  device: { provider: 'emulator', autoDiscover: true }, // or pool: [avd1, avd2, avd3]  
}
```

- Each worker grabs a device and spawns its **own Appium** on a staggered port
- `workers > 1` needs an adequate `pool` or `autoDiscover: true` (else it fails at load)
- `fullyParallel` changes scheduling granularity (tests-in-a-file vs whole files) — it **never** raises the device count
- ⚠ A `pool` hardcodes **AVD ids + serials that only exist on the machine that wrote it** — swap in your own from `npx taqwright devices`, or use `autoDiscover` and stay portable

🔄 Two emulators, one command — the device fan-out is **declarative**, not wiring you hand-roll.



Lab · `tutorial_06_parallel_lab/` ⚡

```
cd tutorial_06_parallel_lab
npm install
npx taqwright test --project android-auto-1      # autoDiscover · serial (1 device)
npx taqwright test --project android-auto-2      # autoDiscover · 2-wide (needs 2 AVDs)
npx taqwright test --project android-single      # ⚠ edit the AVD name/udid first
npx taqwright test --project android-pool-2      # ⚠ edit the 2 pool entries first
```

⚠ `android-single` + `android-pool-2` are pinned to *my* emulators (`Pixel_10_Pro_XL` , `emulator-5554` , ...) — **yours are named differently**. Run `npx taqwright devices` and paste your own `avd:<id>` + serial into `taqwright.config.ts` . The two **auto** projects need no edits and run anywhere.

4 projects, one per strategy: pinned `udid` · `pool` of 2 · `autoDiscover` (1- & 2-wide). Worker *i* gets its own device + Appium on `4723+i` . Adapted from the [taqwright-demo](#) config. → [tutorial_06_parallel_lab/](#)



13 · Parameterize tests

```
const accounts = [
  { label: 'standard', username: 'emma@demoapp.com', password: '10203040' },
  { label: 'admin', username: 'admin@demoapp.com', password: 'admin123' },
];

for (const a of accounts) {
  test(`login as ${a.label}`, async ({ mobile }) => {
    await mobile.getByXPath("//*[@hint='Username']").fill(a.username);
    // ...
  });
}
```

Also: **option fixtures** (`test.extend` + `test.use`), **project-based** (same specs × devices), and **external CSV/JSON** loaded at collection time.  Classic **data-driven** testing, the taqwright way.



Lab · `tutorial_07_parameterize_lab/` 1 2 3 4

```
cd tutorial_07_parameterize_lab
npm install
npx taqwright test --list      # one test per data row
npx taqwright test tests/parameterize.spec.ts
```

Loop a data array → one `test()` per row (valid · wrong password · unknown user). Same flow driven from an external **JSON** file in `from-json.spec.ts`. Slide also mentions option fixtures (`test.extend`) + project-based. →

[tutorial_07_parameterize_lab/](#)



14 · Projects

A **project** = a named group of tests with one **use**: a platform, a device, a build, and its artifacts.

```
projects: [  
  { name: 'android', use: { platform: Platform.ANDROID, device: { /*...*/ } } },  
  { name: 'ios',      use: { platform: Platform.IOS,      device: { /*...*/ } } },  
]
```

```
npx taqwright test --project android --project ios # repeatable
```

Fields: **name** · **use** · **dependencies** · **testMatch** · **testIgnore**. Dependencies run first; if one fails, its dependents **skip**.

Same specs → Android, iOS, BrowserStack.



Lab · `tutorial_08_projects_lab/`

```
cd tutorial_08_projects_lab
npm install
npx taqwright test --project android    # Android login
npx taqwright test --project ios        # iOS login (macOS + simulator)
npx taqwright test                      # both targets
```

Two `projects[]` — an `android` and an `ios` target, each with its own `use` (platform · device · build · bundle id) running its own `login.spec.ts`. Same idea scales to BrowserStack. → [tutorial_08_projects_lab/](#)



15 · Reporters

```
reporter: [  
  ['list'], // one line per test (default)  
  ['html', { open: 'never' }], // browsable folder  
  ['json', { outputFile: 'reports/out.json' }],  
]
```

Reporter	Use
list · line · dot	console — verbose → compact → one char/test
html	self-contained browsable report
json · junit	custom tooling · CI ingest
blob · github	shardable archive · GH Actions annotations

`npx taqwright merge-reports reports/blob --reporter html` combines sharded runs.



16 · Sharding

Split one suite across **machines** for CI speed:

```
npx taqwright test --shard 1/4 --reporter blob    # machine 1 of 4
# ...machines 2-4 → 2/4, 3/4, 4/4
npx taqwright merge-reports ./all-blobs --reporter html
```

- **How tests are assigned:** taqwright sorts every test **deterministically**, then splits into **N** even-by-**count** slices — machine **i** runs slice **i**. Automatic & repeatable; you **can't pin** a test to a machine.
- **Unit:** whole **files** by default; `fullyParallel: true` → individual **tests** (finer balance)
- ⚠ Balances by **count, not duration** (one slow test = a long-pole shard) · each shard needs its **own device**
- GitHub Actions: a **matrix** of 4 jobs upload blobs → a merge job builds one HTML report



tutorial_09_merge_report_lab — how shard + merge fit 🧬

The lab runs **both shards on ONE machine, back-to-back** — a local stand-in for CI's many machines.

```
ONE machine · two slices run sequentially:  
  shard 1/2  --reporter blob → report-1.zip  
  shard 2/2  --reporter blob → report-2.zip  } → all-blobs/ → merge-reports → 1 HTML report
```

- **--shard i/N** runs a *slice* of the suite (file-level); **--reporter blob** emits one archive per slice
- 📦 **because it's one machine**, the blob reporter **clears blob-report/** **between the two runs**
→ each shard **moves** its blob into **all-blobs/** first (otherwise shard 2 overwrites shard 1)
- **In real CI it's N separate machines**: each runs one shard + uploads its blob → a merge job downloads them all → **merge-reports all-blobs --reporter html** → one report



Lab · `tutorial_09_merge_report_lab/`

```
cd tutorial_09_merge_report_lab
npm install
npm run shard1      # --shard 1/2 --reporter blob → all-blobs/
npm run shard2      # --shard 2/2 --reporter blob → all-blobs/
npm run merge        # merge-reports all-blobs --reporter html
npm run report       # one combined HTML report
```

Each shard emits a **blob** archive; **merge-reports** stitches them into one HTML report — the CI sharding pattern (matrix of N jobs → upload blobs → merge job). → [tutorial_09_merge_report_lab/](#)



17 · Retries

```
npx taqwright test --retries 2
```

```
retries: 2, // config (top-level or per-project)
test.describe.configure({ retries: 3 }); // per-group
```

- On failure, taqwright tears down the session and **reruns from scratch** (fresh Appium session)
- Fails then passes → flagged **flaky** (not failed) in `list` / `html`
- `testInfo.retry` (0, 1, 2...) → capture extra diagnostics only on retries
- `test.describe.serial(...)` retries a whole dependent group together



18 · Timeouts

Timeout	Default	Set with
Test (whole test + hooks + fixtures)	60 s	<code>timeout</code> · <code>--timeout</code> · <code>test.setTimeout()</code> · <code>test.slow()</code> (3×)
Expect / action (assertions and actions)	30 s	<code>expectTimeout</code> · per-call <code>{ timeout }</code>

```
test('slow flow', async ({ mobile }) => {
  test.setTimeout(120_000);
  await expect(mobile.getByText('Done')).toBeVisible({ timeout: 10_000 });
});
```

Polls every **200 ms** until actionable. There is **no** separate `actionTimeout` , `navigationTimeout` , or `globalTimeout` .



19 · Trace viewer

Turn on per-action tracing, then scrub the whole run:

```
trace: 'on-failure',    // off | on | on-failure | retain-on-failure (alias)
```

A trace contains: a **clickable timeline** (failures highlighted), a **screenshot after each action**, action name / args / errors, **page-source XML** snapshots, and a network panel.

```
npx taqwright show-report          # → open the test → "taqwright-trace" attachment  
# or directly: test-results/<test-name>/trace.html
```

Self-contained HTML (not a Playwright `.zip`). ~100–300 ms/action overhead → use `on-failure` in CI. Console logs & video are separate artifacts.



Lab · `tutorial_10_retry_timeout_trace_lab/`

```
cd tutorial_10_retry_timeout_trace_lab
npm install
npx taqwright test           # retries: 2 + trace: 'on' (every test traced)
npx taqwright show-report    # open a test → "taqwright-trace" → step the timeline
```

One lab for §17–19: a **flaky** test that clears on retry (flagged *flaky*, not failed) · **timeout** knobs (`test.setTimeout` / per-assertion `{ timeout }`) · a **trace** on every test → scrub it in the report. → [tutorial_10_retry_timeout_trace_lab/](#)



Page Objects — one spec, both platforms

The **flow** is the same on Android & iOS; only the **locators** differ → flow in an **abstract base**, locators in **subclasses**.

```
abstract class LoginPage {                                // one page object per SCREEN
  protected abstract loginButton(): Locator;            // ← subclasses fill these in
  async login(u, p): Promise<HomePage> {                // ...and the flow returns the NEXT screen
    /* ... */ await this.loginButton().click();
    return homePage(this.mobile, this.projectName);
  }
}
class AndroidLoginPage extends LoginPage {
  loginButton() { return this.mobile.getByUiSelector('...description("Login")'); } // no id → UiSelector
}
class IosLoginPage extends LoginPage {
  loginButton() { return this.mobile.getId('Login'); } // iOS accessibility id
}
```

A factory **per screen** picks the subclass by `testInfo.project.name`. Flow methods return the next page, so the spec is a fluent chain — `login()` → `openCatalog()` → `searchAndAddToCart('black')` — **no raw locators, no `if (platform)`**.

One spec → runs on the `android` **and** `ios` projects.



Lab · `tutorial_11_page_object_lab/`

```
cd tutorial_11_page_object_lab
npm install
npm run test:android      # the SAME spec, Android page object
npm run test:ios          # the SAME spec, iOS page object (macOS)
npx taqwright test --list # one spec under [android] AND [ios]
```

One folder per screen under `pages/` — `login/` · `home/` · `search-catalogue/` · `cart/` — each holding its base (flow) + `Android*` / `Ios*` subclasses (locators only), with a factory per screen in `pages/index.ts` . Locators came from real codegen recordings on both platforms. → [tutorial_11_page_object_lab/](#)



Run on BrowserStack — config changes

Same spec, same `mobile` API — only the project's `use` changes. Add a cloud project:

```
{
  name: 'browserstack',
  use: {
    platform: Platform.ANDROID,
    device: { provider: 'browserstack', name: 'Google Pixel 8', osVersion: '14.0' },
    buildPath: 'bs://<app-id>', // YOUR upload - ids are per-account
    appBundleId: 'com.taqelah.demo_app',
    resetBetweenTests: true,
  }, // no local `appium` block - BS runs the server
}
```

- 🔑 Creds via `env` (not config): `BROWSERSTACK_USERNAME` / `BROWSERSTACK_ACCESS_KEY`
- 📦 A `bs://` id is private to the account that uploaded it — upload your own app (`curl -u user:key -X POST .../app-automate/upload -F file=@app.apk`) or point `buildPath` at a local binary
- ☁️ `provider: 'browserstack' + name / osVersion required` · `workers: N` for parallel (no `device.pool`) · vendor knobs via `capabilities → 'bstack:options'`



🔧 Lab · `tutorial_12_cloud_debug_lab` ☁️ — page objects on BrowserStack

`tutorial_11_page_object_lab` 's page objects + spec, verbatim — going cloud changes only `device.provider` .

⚠️ The `bs://...` ids in the config are **mine** — a BrowserStack app id only works for the account that uploaded it. Upload `../app/` 's binaries to **your** account and set `TAQ_APK` / `TAQ_IPA` , or pass a local file path to upload per run.

```
cd tutorial_12_cloud_debug_lab
npm install
cp .env.example .env          # add your BROWSERSTACK_USERNAME / _ACCESS_KEY
                                # ⚠️ + your own TAQ_APK / TAQ_IPA app ids
npm run test:local             # local emulator
npm run test:cloud-android     # BrowserStack — real Android device
npm run test:cloud-ios        # BrowserStack — real iOS device
```

Same spec, same per-screen `pages/` copied from lab 11 — the `browserstack-android` / `browserstack-ios` projects just set `provider: 'browserstack'` + `name / osVersion` ; creds come from `.env` . Runs on real hardware, no local Appium.

Only diff vs lab 11: the factories match `projectName.includes('ios')` , since the cloud project is `browserstack-ios` . →



That's the whole docs tour

Install → codegen → writing tests → running & debugging → **the entire Taqwright Test section**, each with a lab beside it.



Where to go next

```
cd tutorial_01_setup_lab && npm install
npx taqwright doctor           # verify the toolchain
npx taqwright devices          # copy your AVD id into taqwright.config.ts
npx taqwright test             # green? work through the labs in order
```

-  **Every lab folder** has a `README.md` with the full walkthrough
-  Stuck on the `mobile` API, locator priority, config or `lime-cli`? → the **taqwright skill** is the authoritative reference
-  [tutorial_12_cloud_debug_lab/](#) needs a [BrowserStack](#) account



 **Thank you!**

Clone it, run it, break it — the labs are yours to extend. 

★ Star taqwright on GitHub: <https://github.com/taqelah/taqwright>

 Issues & ideas are welcome — on `taqwright` or on this tutorial repo.

Keep testing.

 linkedin.com/in/syam-sasi ·  taqelah.sg